2. Medicare Supplement Cashflow & Anti-Selection Engine — Humana NPV Model (v4.2.0)

Actuarial Overview (Plan G Cohort, Attained-Age Rated, 20-Year Horizon)

From a Humana actuary's perspective, Medicare Supplement is a **long-duration, guaranteed-renewable closed cohort** modeled per 1,000 policies issued at age 65 (240 monthly projection rows):

- **Year-1 Strain:** Year 1 loses money because front-loaded first-year commissions and upfront acquisition costs (underwriting, issue, marketing) exceed first-year underwriting margins.
- **Value Drivers:** Lifetime value is earned in renewal years (Years 2–20) and depends on persistency, keeping annual rate increases aligned with medical trend, and managing **anti-selective shock lapse** on policy anniversaries.

Visual Cashflow & Anti-Selection Architecture

```
I=============================================================================================
==============
                      MEDICARE SUPPLEMENT: 20-YEAR (240-MONTH) CASHFLOW ENGINE
=============================================================================================
==============

 [ISSUE: t = 0] ──▶ Outflow: −1,000 × MS_Acq (Underwriting, Issue, Marketing)
    │
    ▼
  [TWO-CLASS ANTI-SELECTIVE DECREMENT ENGINE (Each Month k = 0 ... 239)]
  Cohort splits at issue into:
    • "Shoppers" (Share = MS_ShopShare): Healthier lives (Relative Cost MS_ShopCost =
0.85 < 1.0)
      who can pass medical underwriting elsewhere if rates spike.
    • "Non-Shoppers" (1 - MS_ShopShare): Impaired/loyal lives with higher relative cost:
      MS_RelN = (1 - MS_ShopShare × MS_ShopCost) ÷ (1 - MS_ShopShare)  [So Year-1 blend =
1.000]
    │
    ├──▶ 1. Mortality (Mid-Month): Gompertz curve q(age) = min(0.35, q65 × e^(MortG × (Age − 65)))
    │      Exposure_Shop = Shop_Start − 0.5 × Deaths_Shop
    │      Exposure_Non  = Non_Start  − 0.5 × Deaths_Non
    │
    └──▶ 2. Dynamic Anniversary Shock Lapse (End of Month 1 of Policy Years 2+):
```

```
                When rate increase hits on anniversary, extra shock lapse triggers:
               Shock_yr = max(0, MS_Shock + MS_ShockSlope × (RateInc - MS_ShockRef))
                Shoppers are MS_ShopProp times as likely to take the shock lapse as
Non-Shoppers!
               Result: Shoppers exit faster → Remaining pool's Claim Relativity drifts UP
(> 1.0).

  [MONTHLY CASHFLOWS]
  (+) Premium (Start of Month):
      Total_InForce × [MS_Prem1 × (1 + MS_PremAge)^(Yr-1) × (1 + MS_RateInc)^(Yr-1)] ÷ 12
  (-) Commissions & Tax (Start of Month):
      Premium × [IF(Yr=1, MS_FYComm, MS_RenComm) + MS_PremTax]
  (-) Incurred Claims (Mid-Month, Class-Weighted):
      [Exp_Shop × MS_RelS + Exp_Non × MS_RelN]
      × [MS_Clm1 × (1 + MS_ClmAge)^(Yr-1) × (1 + MS_Trend)^(Yr-1) × SelectFactor(Yr)] ÷ 12
  (-) Maintenance Admin (Mid-Month):
      Total_Exposure × [MS_Maint × (1 + ExpInfl)^(Yr-1)] ÷ 12
  (-) IRC §848 DAC Tax (Profit Test):
      9.2% of Net Premiums capitalized & amortized over 180 mos (-$101,388 DE NPV impact).
================================================================================
==============
\
```

Step-by-Step Mathematical Equations (MedSupp_Monthly Columns A to AG)

Let month index k run from 0 to 239:

- **Policy Year (Col B):** Year(k) = INT(k ÷ 12) + 1
- **Month of Policy Year (Col C):** MonthOfYr(k) = (k mod 12) + 1
- **Attained Age (Col D):** Age(k) = MS_IssueAge + Year(k) − 1

1. Gompertz Mortality Decrement (Col E)

Annual mortality follows a Gompertz law anchored at age 65 and capped at 35% per year, converted to a compound monthly rate:

- **Annual Death Rate:**
  q_death_annual(Age) = min(0.35, MS_Q65 × exp(MS_MortG × (Age − 65)))
- **Monthly Death Rate:**
  q_death_monthly(k) = 1 − (1 − q_death_annual(Age))^(1/12)

2. Base Durational Lapse & Dynamic Anniversary Shock Lapse (Cols F–J)

- **Base Annual Lapse (Col F):**
  Lapse_base_yr(k) = min(1.0, MS_Lapse(min(Year(k), 3)) × MS_LapseMult)
- **Dynamic Shock Lapse in Policy Years 2+ (Col G):**
  When F_DynLapse = 1, shock lapse scales linearly with the excess of the filed rate increase over the reference rate increase:
  Shock_yr(k) = max(0, MS_Shock + MS_ShockSlope × (MS_RateInc + MS_RateAdd − MS_ShockRef)) for Year(k) ≥ 2 (and 0 for Year(k) = 1)
- **Total Annual Lapse (Col H):**
  Lapse_total_yr(k) = min(1.0, Lapse_base_yr(k) + Shock_yr(k))
- **Base Monthly Lapse Rate (Col J):**
  q_lapse_base_mo(k) = 1 − (1 − Lapse_base_yr(k))^(1/12)
- **Block Monthly Lapse Rate with Month-1 Anniversary Shock (Col I):**
  Because the rate increase hits on the policy anniversary (MonthOfYr(k) = 1), all of the year's extra shock lapse occurs in Month 1 of the policy year, solved so that Month 1 survival times 11 months of base survival equals exact annual survival (1 − Lapse_total_yr(k)):
- • If MonthOfYr(k) = 1:
  q_lapse_block_mo(k) = 1 − (1 − Lapse_total_yr(k)) ÷ (1 − Lapse_base_yr(k))^(11/12)
- • If MonthOfYr(k) > 1:
  q_lapse_block_mo(k) = q_lapse_base_mo(k)

3. Two-Class Anti-Selection Roll-Forward (Shoppers vs. Non-Shoppers, Cols K–T)

At issue (k = 0), the cohort MS_Cohort (1,000 policies) is partitioned into:

- **Shoppers at Issue (Col K, k = 0):** Shop(0) = MS_Cohort × MS_ShareEff
- **Non-Shoppers at Issue (Col L, k = 0):** Non(0) = MS_Cohort × (1 − MS_ShareEff)
- **Initial Claim Relativities:**
  • Shopper Claim Relativity: MS_RelS = MS_ShopCost (0.85 in base case)
  • Non-Shopper Claim Relativity: MS_RelN = (1 − MS_ShopShare × MS_ShopCost) ÷ (1 − MS_ShopShare)
  (Calibrated so the issue-mix average relativity is MS_ShopShare × MS_RelS + (1 − MS_ShopShare) × MS_RelN = 1.0000.)

In each month k:

- **Deaths (Mid-Month, Cols M–N):**
  Deaths_Shop(k) = Shop(k) × q_death_monthly(k)
  Deaths_Non(k) = Non(k) × q_death_monthly(k)

- **Survivors of Death:**
  Surv_Shop(k) = Shop(k) − Deaths_Shop(k)
  Surv_Non(k) = Non(k) − Deaths_Non(k)
- **Extra Shock Lapse Split Between Classes (Cols O–P):**
  Both classes experience the base monthly lapse q_lapse_base_mo(k). The extra shock lapse [q_lapse_block_mo(k) − q_lapse_base_mo(k)] is allocated so a Shopper has MS_PropEff times the extra lapse rate x(k) of a Non-Shopper while preserving the exact total block lapse count:
  x(k) = [(Surv_Shop(k) + Surv_Non(k)) × (q_lapse_block_mo(k) − q_lapse_base_mo(k))] ÷ [MS_PropEff × Surv_Shop(k) + Surv_Non(k)]
- • Non-Shopper Lapses (Col O): Lapses_Non(k) = Surv_Non(k) × [q_lapse_base_mo(k) + x(k)]
- • Shopper Lapses (Col P): Lapses_Shop(k) = Surv_Shop(k) × min(1.0, q_lapse_base_mo(k) + MS_PropEff × x(k))
- **Next Month In-Force Roll-Forward:**
  Shop(k + 1) = Shop(k) − Deaths_Shop(k) − Lapses_Shop(k)
  Non(k + 1) = Non(k) − Deaths_Non(k) − Lapses_Non(k)
- **Total In-Force & Mid-Month Exposure (Cols Q–T):**
  InForce(k) = Shop(k) + Non(k)
  Deaths(k) = Deaths_Shop(k) + Deaths_Non(k)
  Exposure(k) = InForce(k) − 0.5 × Deaths(k)

4. Monthly Cashflows & Adverse Selection Drift (Cols U–AG)

- **Annualized Premium per Policy (Col U):**
  PremRate(k) = MS_Prem1 × (1 + MS_PremAge)^(Year(k) − 1) × (1 + MS_RateInc + MS_RateAdd)^(Year(k) − 1)
- **Annualized Issue-Mix Claim Cost per Policy (Col V):**
  ClmRate(k) = MS_Clm1 × (1 + MS_ClmAge)^(Year(k) − 1) × (1 + MS_Trend + MS_TrendAdd)^(Year(k) − 1) × SelectFactor(min(Year(k), 3))
- **Monthly Premium Billed in Advance (Col W):**
  Premium(k) = InForce(k) × PremRate(k) ÷ 12
- **Monthly Class-Weighted Incurred Claims (Col X):**
  Claims(k) = [(Shop(k) − 0.5 × Deaths_Shop(k)) × MS_RelS + (Non(k) − 0.5 × Deaths_Non(k)) × MS_RelN] × ClmRate(k) ÷ 12
- **In-Force Block Claim Relativity Drift (Col AG):**
  BlockRelativity(k) = Claims(k) ÷ [Exposure(k) × ClmRate(k) ÷ 12]
  (Because Shoppers exit faster at every anniversary rate increase, BlockRelativity(k) starts at 1.0000 in Month 1 and drifts monotonically upward over the 20-year projection.)

Key Valuation Results & Portfolio Insights (Approved Set AS-SYN-2026-009)

Metric / Segment	Cash NPV	DE NPV (After Tax & Capital)	Monte Carlo Mean (Cash / DE)	P(NPV < 0) (Cash / DE)	PV Loss Ratio	Break-Even Year
Plan G at Age 65 (Headline Cell, per 1,000)	**$692,009**	**$411,854**	$688,437 / $408,165	0.6% / 3.0%	75.7%	Year 2 / Year 2
Full Med Supp Portfolio (Issue-Age x Plan Mix)	**$310,389**	**$119,341**	—	—	78.2%	—
Attained-Age Rating Basis (Portfolio)	**$310,389**	—	—	—	78.2%	—
Issue-Age Rating Basis (GA, AZ, FL, MO)	**$276,496**	—	—	—	78.2%	—
Community Rating Basis	**$277,967**	—	—	—	78.2%	—

Critical Actuarial Takeaways for the Interview

1. **A Headline Cell is Not a Portfolio:** While Plan G at age 65 earns $411,854 DE NPV per 1,000 policies, the realistic sales mix across issue ages and plans earns $119,341 DE NPV (29% of headline). High-Deductible Plan G (HDG) and issue ages 75+ lose money because fixed per-policy maintenance costs do not scale down with small HDG premiums, and filed attained-age premium steps (3.0%) are flatter than the underlying age morbidity slope (3.8%).
2. **Management Rate-Filing Response Cuts Tail Risk:** In the Monte Carlo simulation, without a dynamic rate-filing response (gap closure = 0), P(Cash NPV < 0) is 18.6%. When future rate filings respond to emerging claims with a 2-year lag (a filing effective in year s+1 is

prepared from experience through year s−1), P(Cash NPV < 0) drops to 0.6% (and P(DE NPV < 0) to 3.0%).

3. **Break-Even Cushion:** Break-even DE claim trend is 7.44% (+0.94 pt of headroom above the 6.50% base trend), and break-even rate increase is 6.32%.
4. **v4.2 Risk-Based Pricing Recommendation:** Priced so that P(DE NPV < 0) ≤ 10%. Plan G at 65 meets the target at the current $1,980 premium with 1.5% headroom (P(DE < 0) = 3.0%), but only because rate filings keep responding to emerging claims: with no filing response it is 24.7%.
5. **Portfolio Repricing by Cell:** The age × plan portfolio fails the target as priced (P(DE < 0) = 28.0%). Repricing Plan N +2.0% to +8.1% and older-age Plan G +1.3% to +6.6% brings it to 3.1% (DE NPV $382,991 per 1,000). High-Deductible G cannot reach the target margin inside the 65% minimum loss ratio, so it is reported as capped rather than priced (output/npv/medsupp_cell_pricing.csv).